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جامعة الإمام عبد وزارة التعليم
كلية علوم الرحمن بن فيصل
الحاسب وتقنية المعلومات

جامعة الإمام عبدالرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

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Abstract

This project focuses on designing a high-fidelity prototype of an Explainable Artificial Intelligence (XAI) helper system for visually impaired users in Saudi Arabia. The prototype is connected to the Sehhaty healthcare application and aims to make booking medical appointments easier and more accessible. Many visually impaired users face difficulties when using healthcare applications because some interfaces are complicated, unclear, and not fully designed for accessibility needs.

To help solve this issue, our prototype provides simple step-by-step guidance during the appointment booking process. The system includes clear navigation, accessible interface elements, audio support, and confirmation messages to help users complete tasks more confidently and independently. Users can select a clinic, choose a date and time, review their appointment details, and confirm the booking in a simple and organized way.

The prototype was designed using Human-Computer Interaction (HCI) principles such as reducing cognitive load, supporting user understanding, and increasing trust in the system through clear feedback and explanations. Overall, the project aims to improve accessibility and create a more comfortable and inclusive digital healthcare experience for visually impaired users.

1.Introduction

Artificial Intelligence (AI) technology usability has increased in recent years and become an important tool in many fields such as healthcare, education, digital governance services. AI can analysis large amounts of data and process these data faster and in less time compared to humans. The cons are that AI-based systems lack clarity in that people with disabilities have difficulty understanding how they help. To improve clarity of use for disabled people the concept of Explainable Artificial Intelligence (XAI) has introduced to provide more clarity, understanding, trust and transparency of AI systems.

This project focus on designing and developing an Explainable AI (XAI) digital helper system for people with visual impairment in Saudi Arabia specially in healthcare field. Accessing these digital services is a big challenge for visual impairment because of the complexity of the interfaces, and the lack of task description during task completion. The most popular example is booking a medical appointment through government application such as Sehaty, that requires a lot of steps that might not be easy to accomplish.

To overcome these accessibility challenges for visually impaired users, we developed CarePoint. CarePoint is a prototype healthcare application designed to significantly improve the user experience by integrating an Explainable AI (XAI) helper. This digital assistant guides the user through every phase of the application, from the initial sign-in process to selecting a healthcare service and receiving final confirmation. By providing clear step-by-step transparency, CarePoint ensures that users can navigate the platform independently and with complete confidence.

Proposed system aims to make digital healthcare services easier and more comfortable for visually impaired users by integrating Explainable Artificial Intelligence (XAI) into healthcare applications. Simple guidance, clear explanations, and supportive assistance will enable users to perform healthcare tasks with more confidence and independence. In this way, the system will contribute to creating a more accessible and user-friendly healthcare experience in Saudi Arabia.

2.Literature Review

Artificial Intelligence (AI) technologies are becoming more common in healthcare and digital services. Many systems now use AI to help users complete tasks more efficiently and improve accessibility. However, visually impaired users still face difficulties when using digital healthcare applications because many interfaces depend mainly on visual interaction. Complex navigation, unclear instructions, and lack of accessibility support can make healthcare tasks difficult to complete independently. Explainable Artificial Intelligence (XAI) is one of the technologies that improves user understanding by providing clear explanations and guidance during system interaction. Several previous studies discussed how AI and accessibility-focused systems can support visually impaired users and improve digital usability.

Study 1: Web Accessibility and Explainable AI

Kesharia et al. (2024) developed a web accessibility system for visually impaired users using speech recognition and text-to-speech technologies. The system allowed users to navigate websites through voice commands while receiving spoken explanations for interface elements and interaction steps. The study showed that users were able to understand website structures and complete tasks more easily when clear audio guidance was provided. This study is related to the CarePoint project because both systems focus on helping visually impaired users interact with digital systems using explainable guidance and accessible interaction methods.

Study 2: Accessible Explainable Artificial Intelligence Systems

Nwokoye et al. (2024) reviewed different accessible Explainable Artificial Intelligence systems and examined how explanations are presented to users with disabilities. The researchers found that many systems still depend heavily on visual explanation methods such as charts and graphs, which are difficult for visually impaired users to understand. The study recommended presenting explanations in different formats including audio support and simplified text to improve accessibility and usability. These findings support the goals of CarePoint because the proposed system also focuses on clear explanations and audio guidance during healthcare appointment booking.

Study 3: AI Technologies for Digital Accessibility

Bahtyarovna (2025) explored several AI technologies used to improve digital accessibility for blind users. The study discussed voice assistants, conversational AI systems, and intelligent screen readers that help users interact with digital platforms through natural language communication. The research showed that conversational AI systems improved user independence and helped visually impaired users complete tasks more comfortably. This study supports the idea behind CarePoint because the proposed system also uses supportive AI interaction to guide users through healthcare services.

Study 4: AI and Healthcare Accessibility in Saudi Arabia

Alanazi et al. (2025) studied the role of Artificial Intelligence in improving healthcare accessibility for people with disabilities in Saudi Arabia. The research found that AI-supported healthcare systems improved access to healthcare information and increased user independence during digital interaction. However, the study also identified challenges related to complicated healthcare interfaces and limited accessibility support. This study is highly relevant to CarePoint because it focuses on healthcare accessibility in Saudi Arabia and highlights the need for more accessible healthcare applications for users with disabilities.

Study 5: AI-Assisted Mobility for Visually Impaired Users

Kamran et al. (2024) developed an AI-assisted mobility application that supports visually impaired users through object detection and audio feedback technologies. The application provided spoken descriptions of surrounding objects to help users navigate environments more safely and independently. The study showed that continuous audio guidance improved user confidence and awareness while using the system. This research highlights the importance of audio support and guidance, which are important features included in the CarePoint prototype.

Study 6: Understanding AI Explanations and User Trust

Labarta et al. (2025) introduced a cognitive model called CUE to study how users understand AI-generated explanations. The researchers compared different explanation styles and found that users understood systems more effectively when explanations were simple, organized, and easy to follow. The study also showed that unclear explanations reduced trust and increased confusion during system interaction. This study supports the CarePoint design because the proposed system focuses on providing simple step-by-step explanations and confirmation feedback to improve user understanding and trust.

Overall, previous studies show that AI technologies and accessibility-focused systems can improve digital experiences for visually impaired users. Research highlights the importance of clear explanations, audio guidance, simple navigation, and user-friendly interfaces in improving accessibility and usability. These findings support the development of CarePoint as an Explainable Artificial Intelligence healthcare assistant that helps visually impaired users interact with the Sehhaty application more easily and independently.

3.Problem Statement

Although digital healthcare services in Saudi Arabia have rapidly expanded, many healthcare applications such as Sehhaty are still difficult for visually impaired users to use independently. Important tasks like booking a medical appointment often require users to go through multiple screens, buttons, and options that mainly depend on visual interaction, making the process challenging and sometimes frustrating.

Another issue is that many healthcare applications do not fully support assistive technologies such as screen readers, which can make navigation unclear and confusing. Users may struggle to understand where they are in the process, what step comes next, or whether an action has been completed successfully. In addition, the lack of clear explanations and feedback during task completion can reduce users’ confidence and trust in the system.

To address these challenges, this project proposes CarePoint, an Explainable Artificial Intelligence (XAI) healthcare assistant designed to support visually impaired users while accessing digital healthcare services. The system provides clear step-by-step guidance, audio assistance, simple navigation, and transparent feedback to help users complete healthcare tasks more easily, confidently, and independently.

4.Proposed Solution

The proposed solution is an Explainable Artificial Intelligence (XAI) assistant integrated into the Sehhaty application to support visually impaired users while booking medical appointments. The main goal of the system is to make the booking process easier, clearer, and more accessible for users who may struggle with complicated digital interfaces.

The prototype was designed in Axure as a high-fidelity interface that guides users step-by-step through the appointment booking process. Users can choose a clinic, select the appointment date and time, review their booking details, and confirm the appointment using clear and simple screens. The design also includes large buttons, readable text, simple navigation, audio guidance, and confirmation messages to improve usability and accessibility.

To make the experience more comfortable for users, the prototype applies several HCI principles such as reducing cognitive load by keeping the interface clean and simple, supporting users' mental models through consistent navigation, and increasing trust in automation by providing immediate feedback and clear confirmation screens. Overall, the system aims to help visually impaired users access healthcare services more easily and independently.

5.High Fidelity Prototype

1.1. Get Started Interface

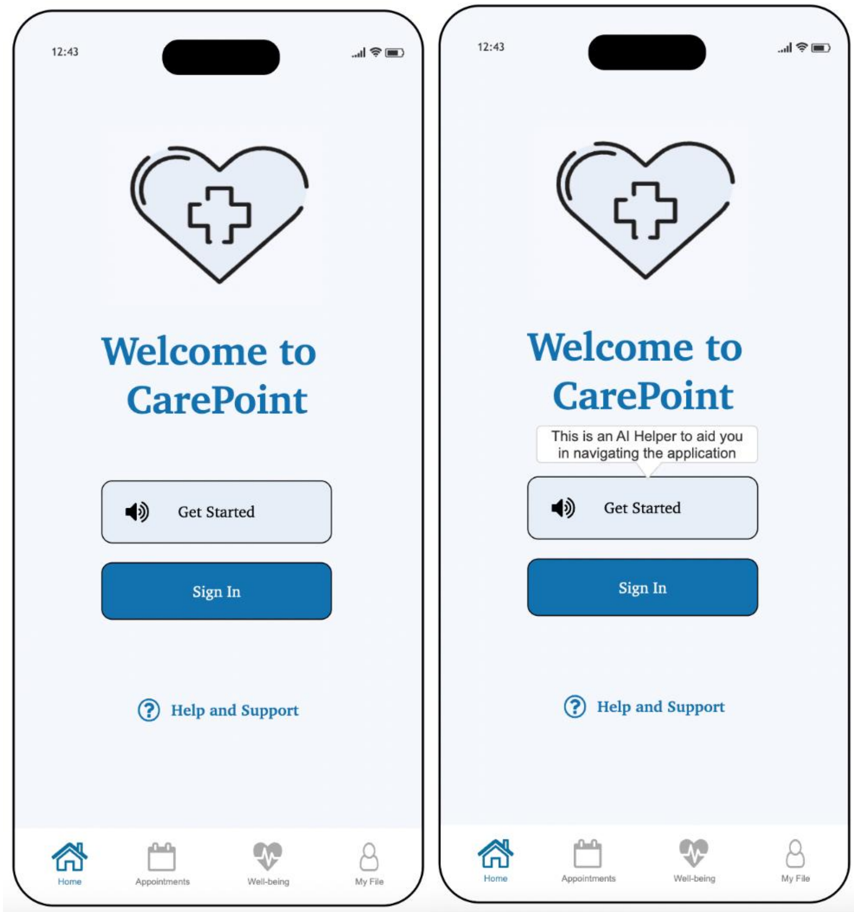

Figure 1 Get Started Interface

1.2. Sign-In Interface

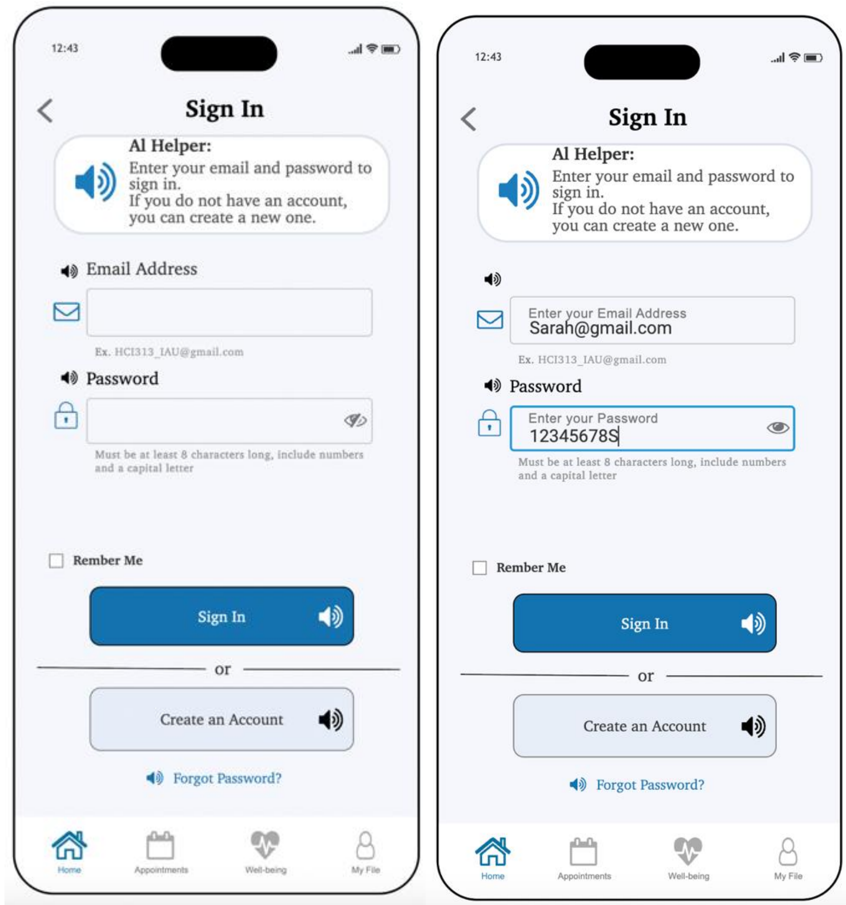

Figure 2. Sign-In Interface

1.2.1. Sign-In Error Message

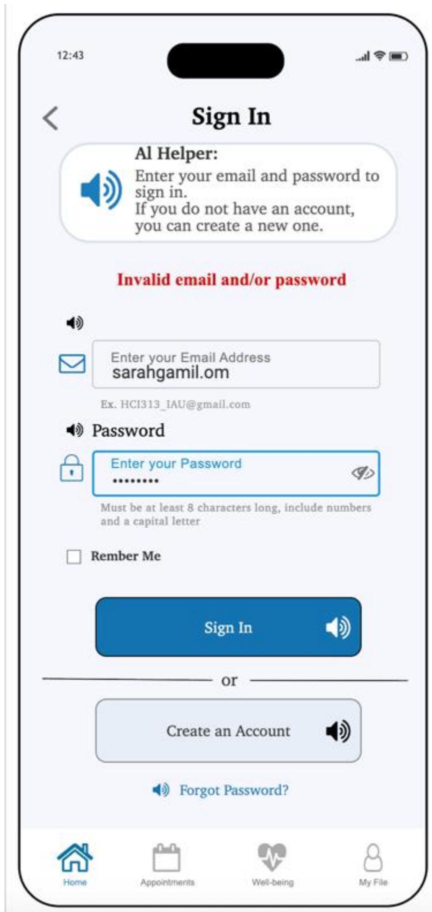

Figure 3. Sign-In Error Message

Note: correct credentials for logging in are shown in the Sign-In Interface

1.3. Create Account Interface

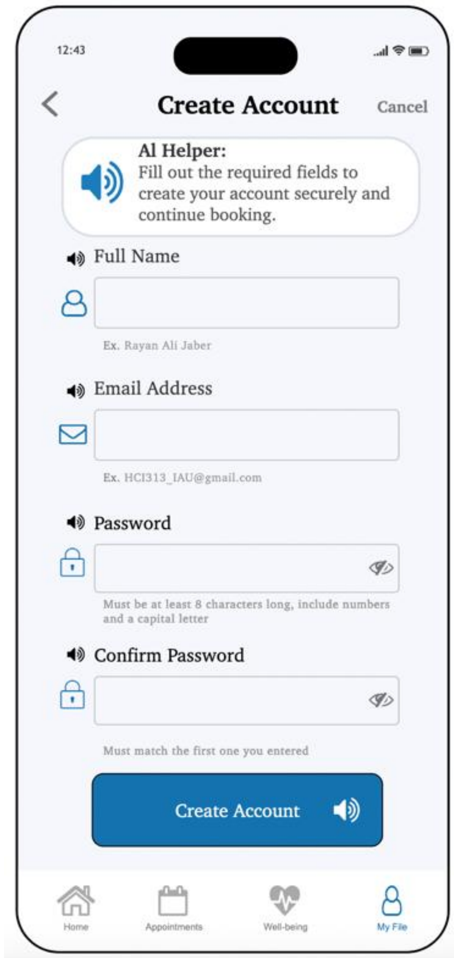

Figure 4 Create Account Interface

1.4. Forgot Password Interface

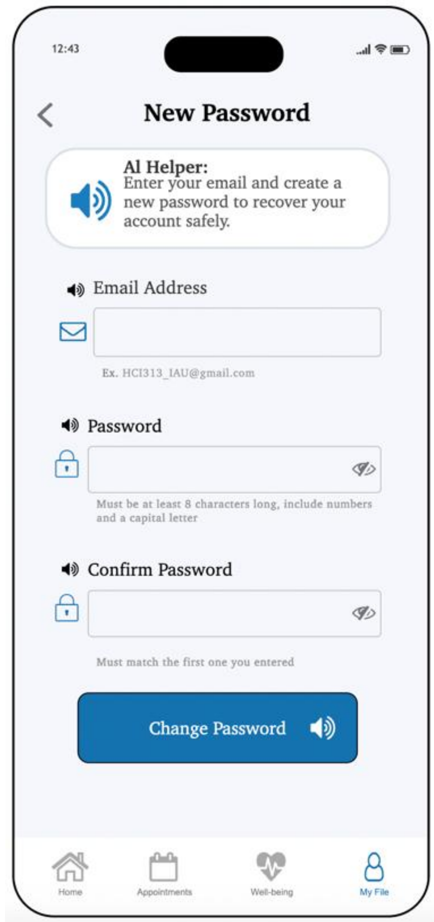

Figure 5. Forgot Password Interface

1.5. Homepage Interface

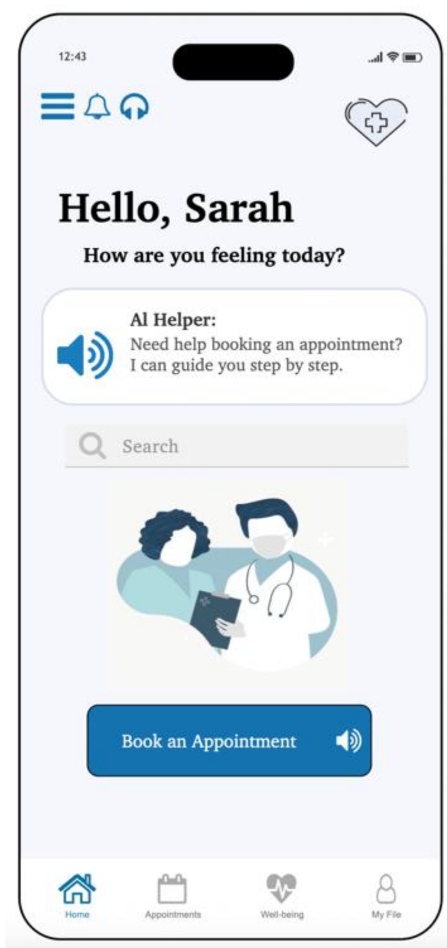

Figure 6 Homepage Interface

1.6. Appointment Type Interface

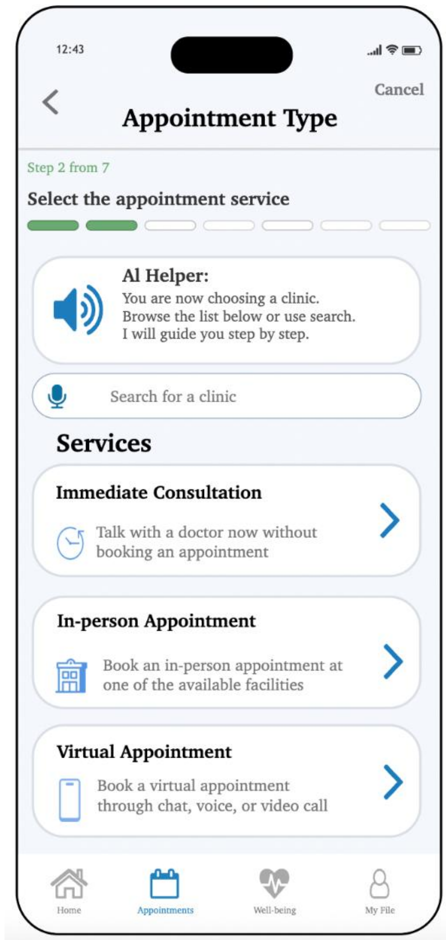

Figure 7. Select Appointment Type Interface

1.7. Immediate Consultation Interface

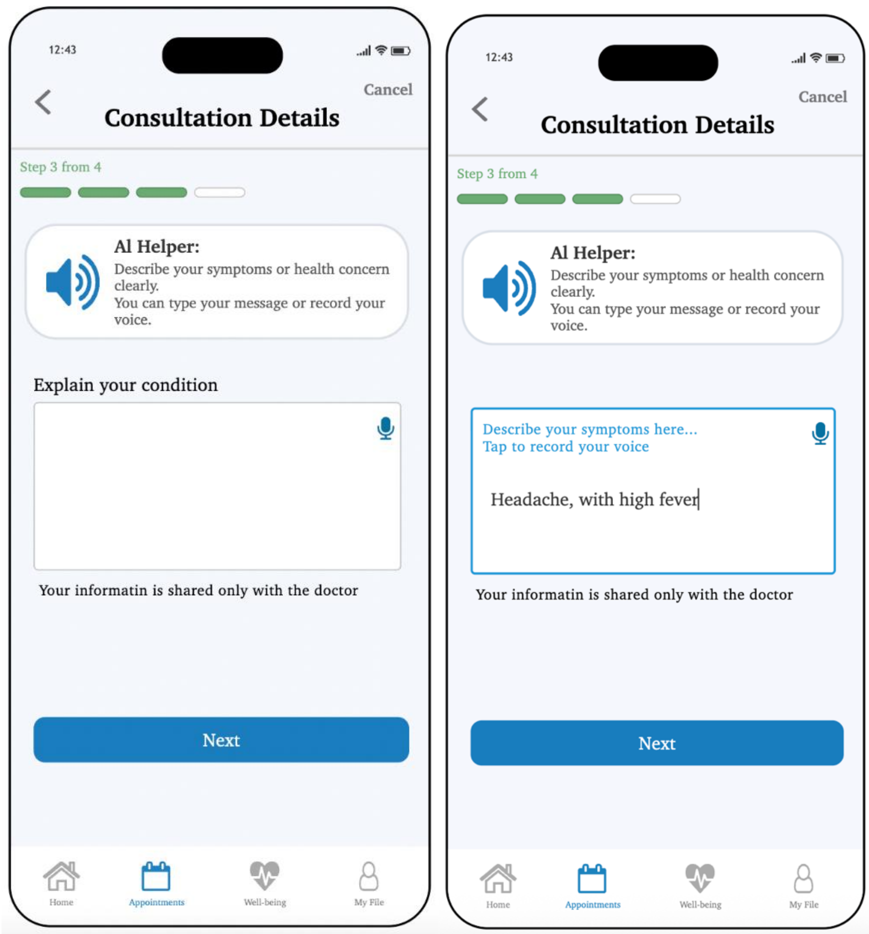

Figure 8. Immediate Consultation Interface

1.8. Confirmation Message Interface

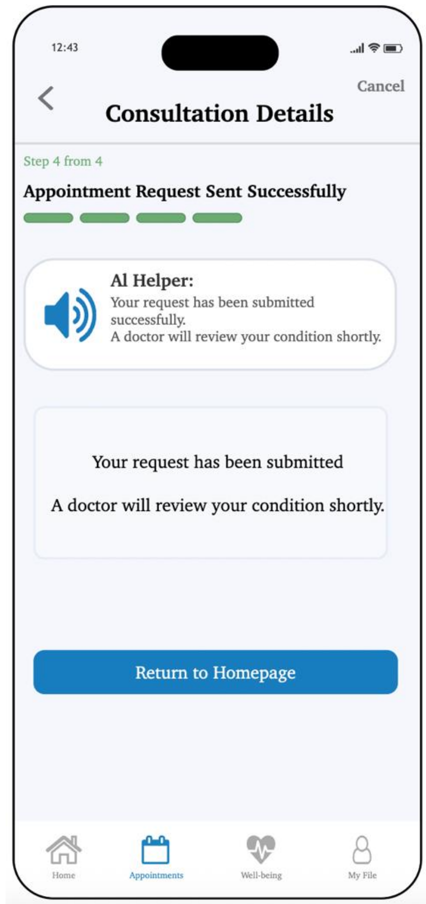

Figure 9. Message Confirmation Interface

1.9. Select a Clinic – In-Person Appointment Interface

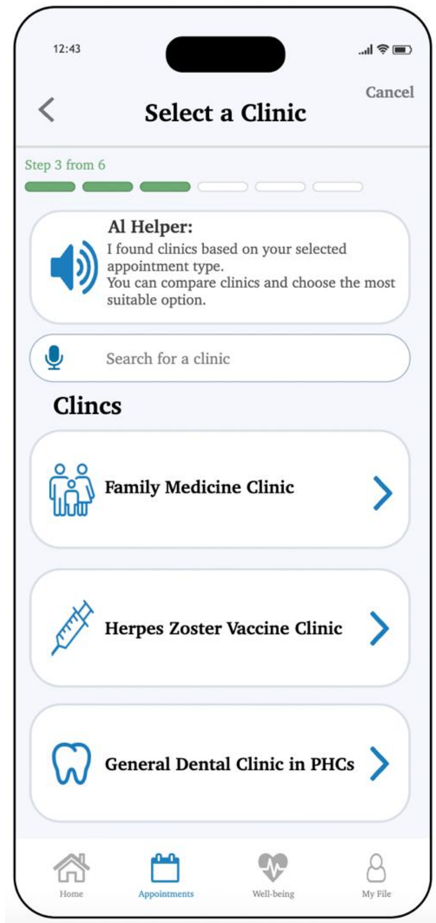

Figure 10. Select a Clinic for an In-person Appointment Interface

1.10. Clinic Location – Nearest Distance Interface

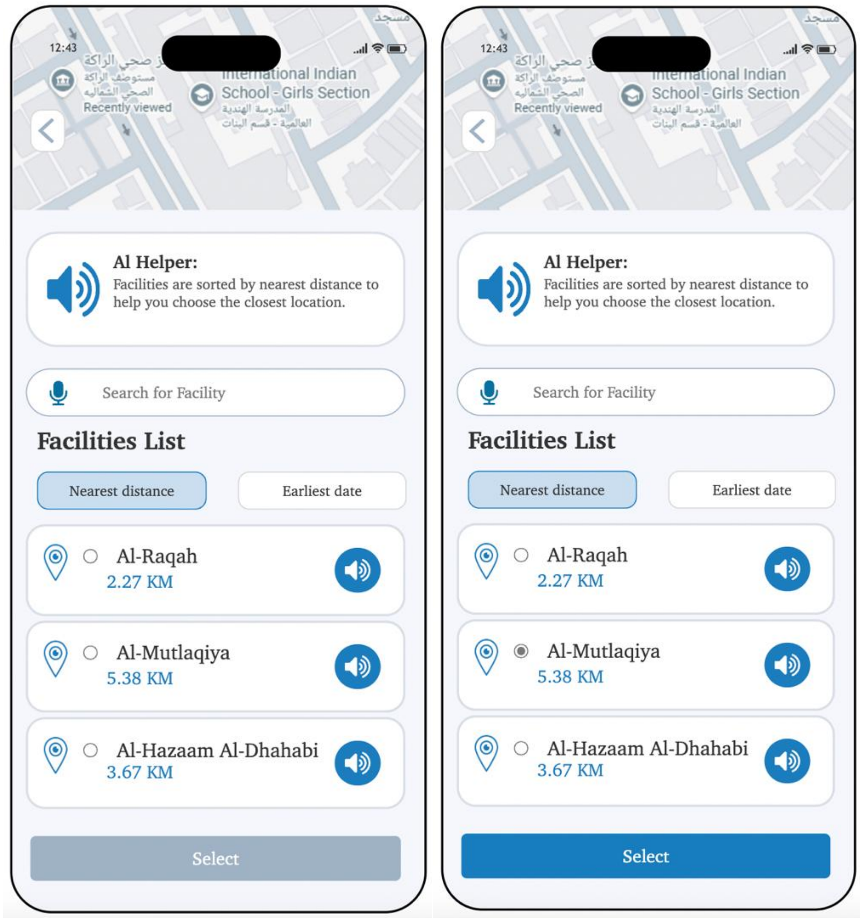

Figure 11.Clinic Location Interface, based on Nearest Distance

1.11. Clinic Location – Earliest Availability Interface

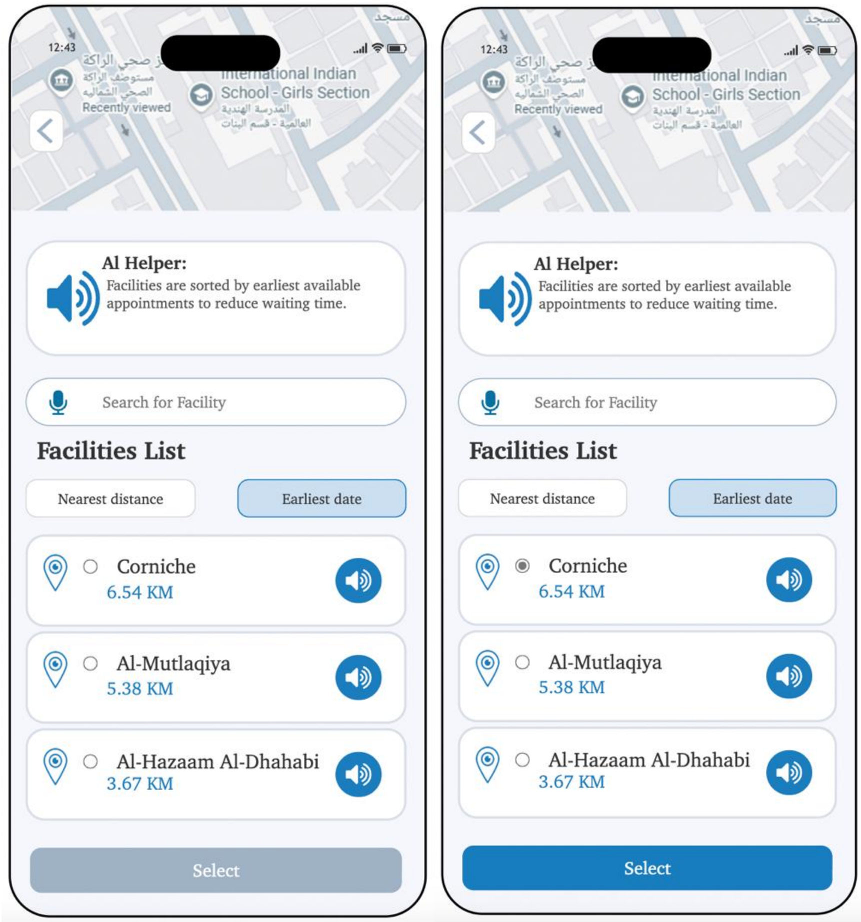

Figure 12. Clinic Location Interface, based on earliest availability

1.12. Select Doctor - In-Person Appointment Interface

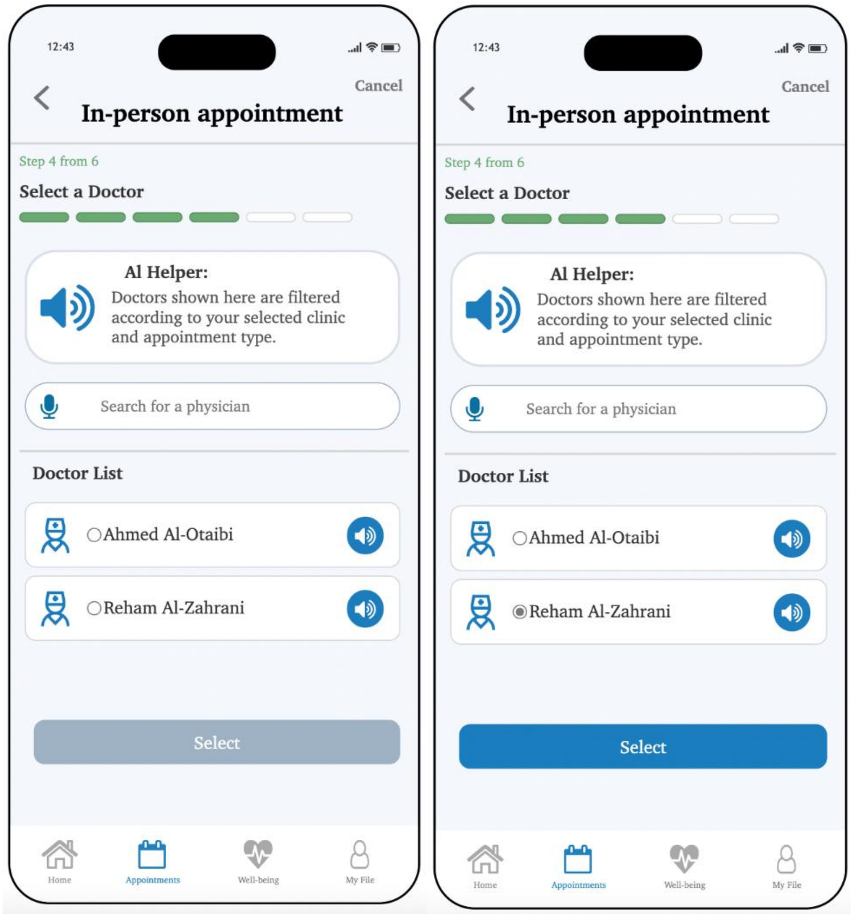

Figure 13. Select a Doctor for In-Person Appointment Interface

1.13. Date & Time - In-Person Appointment Interface

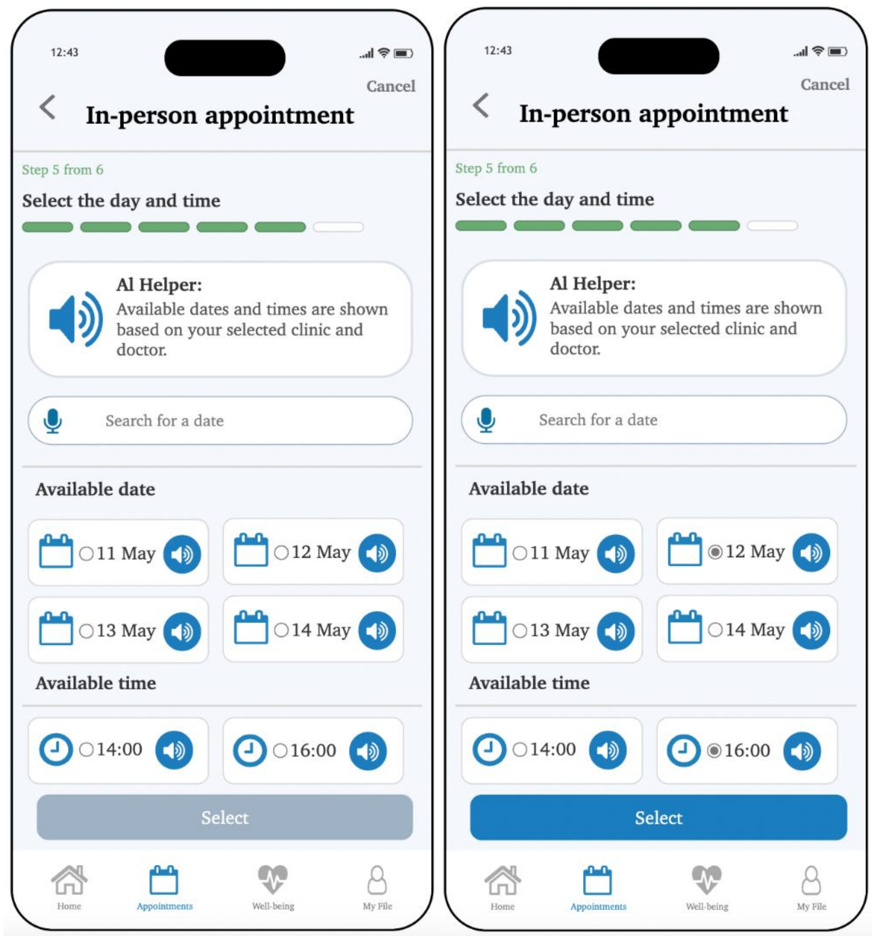

Figure 14. Select Date and Time for In-Person Appointment Interface

1.14. In-Person Appointment Details Interface

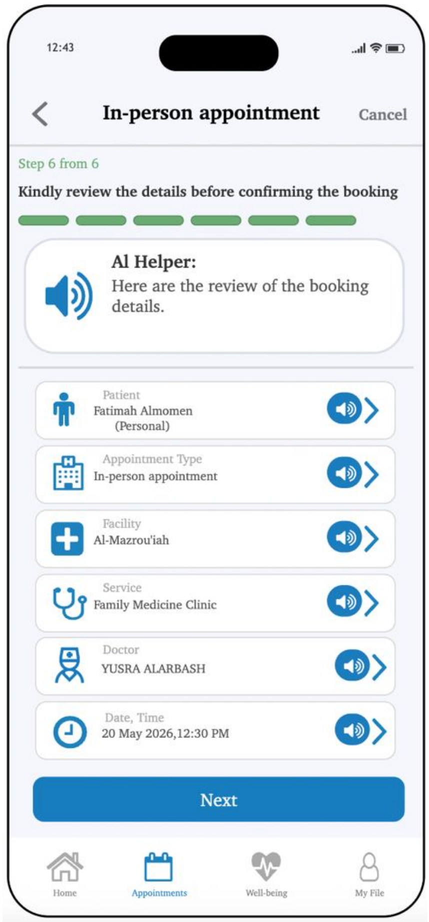

Figure 15. Review In-Person Appointment Details Interface

1.15. Confirm In-Person Appointment Interface

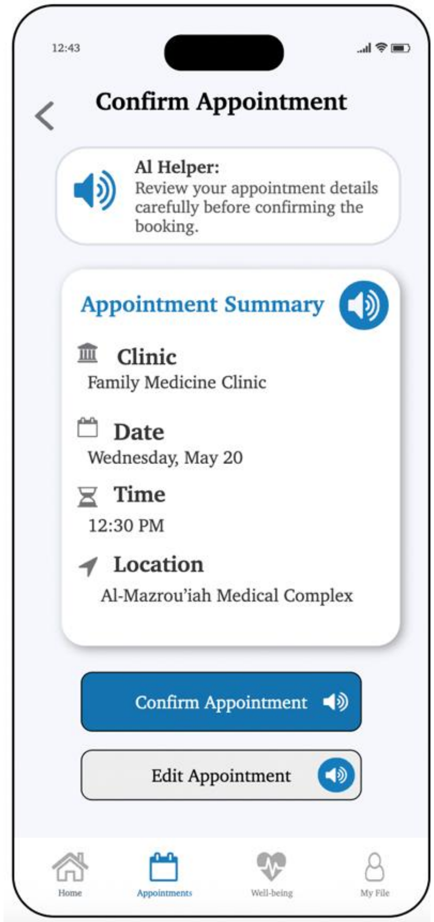

Figure 16. Confirm In-Person Appointment Booking Interface

1.16. Select a Clinic – Virtual Appointment Interface

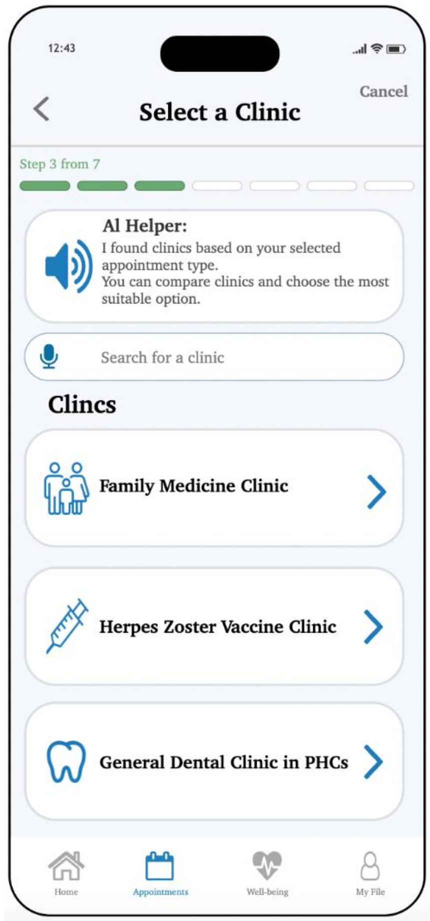

Figure 17. Select a Clinic for a Virtual Appointment Interface

1.17. Virtual Appointment - Communication Method Interface

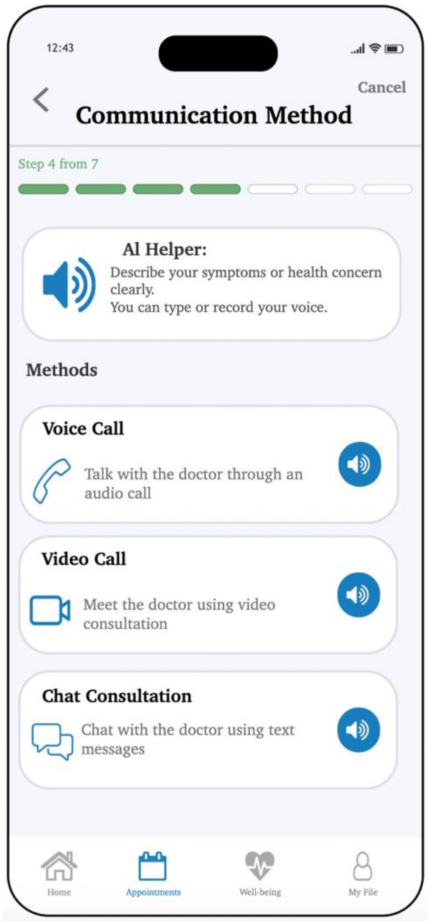

Figure 18. Select a Communication Method for Virtual Appointment Interface

1.18. Select Doctor - Virtual Appointment Interface

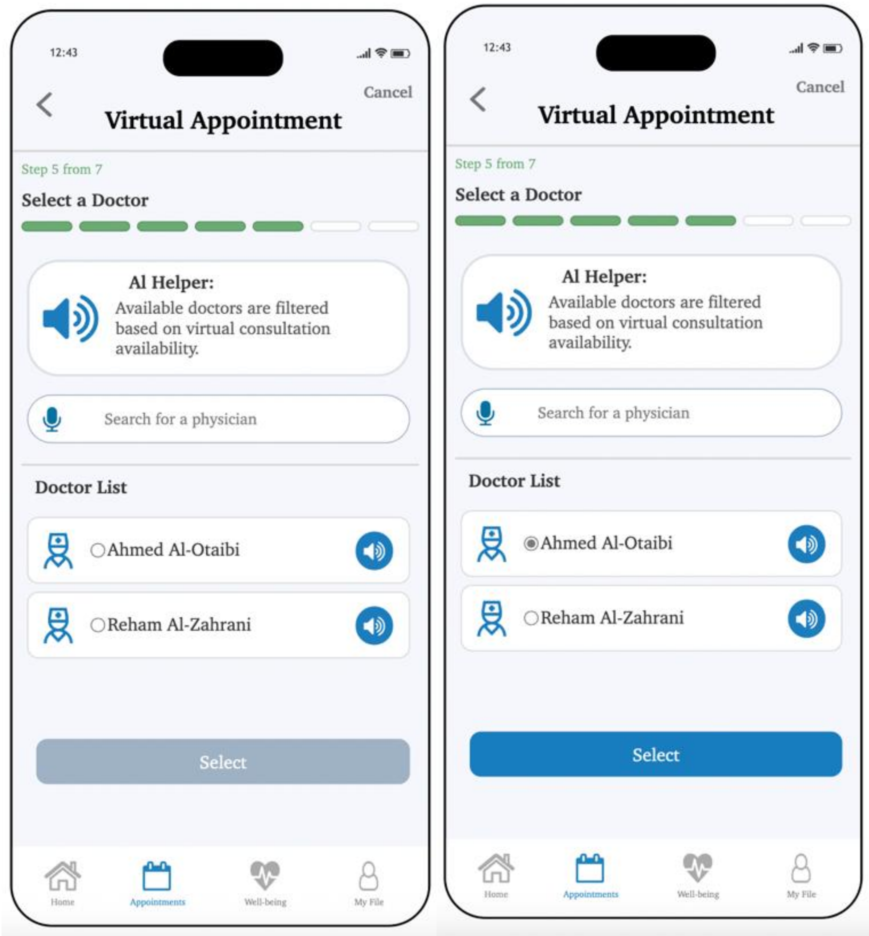

Figure 19. Select a Doctor for Virtual Appointment Interface

1.19. Date & Time - Virtual Appointment Interface

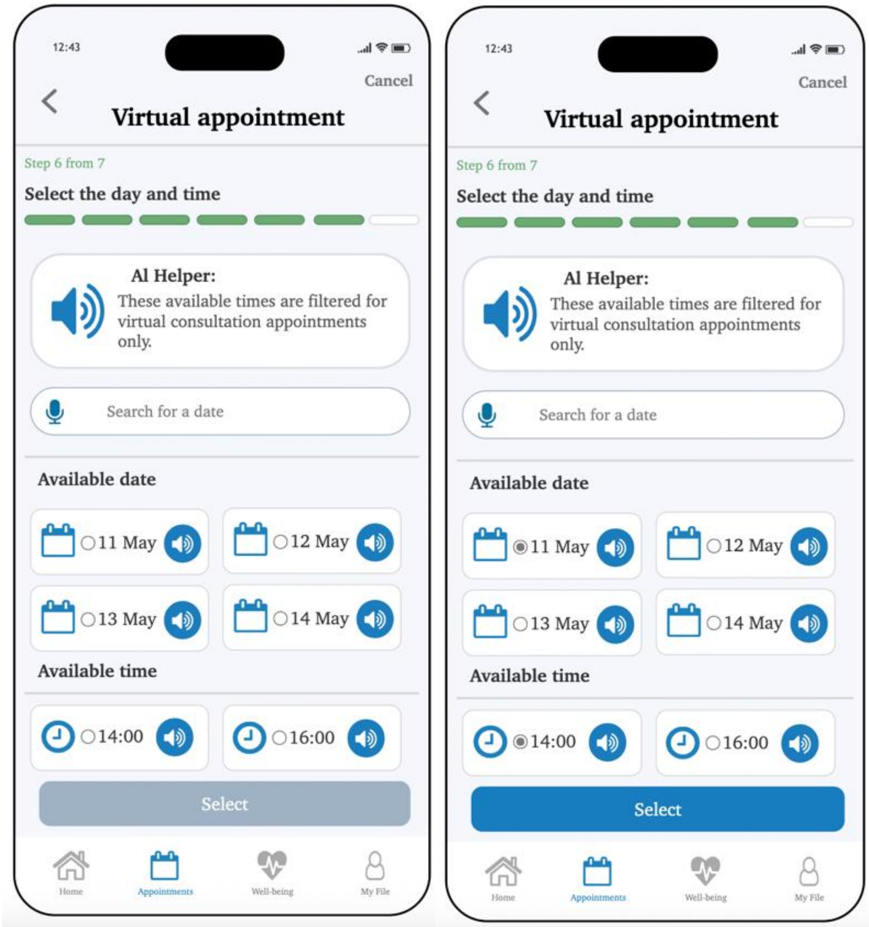

Figure 20. Select Date and Time for Virtual Appointment Interface

1.20. Virtual Appointment Details Interface

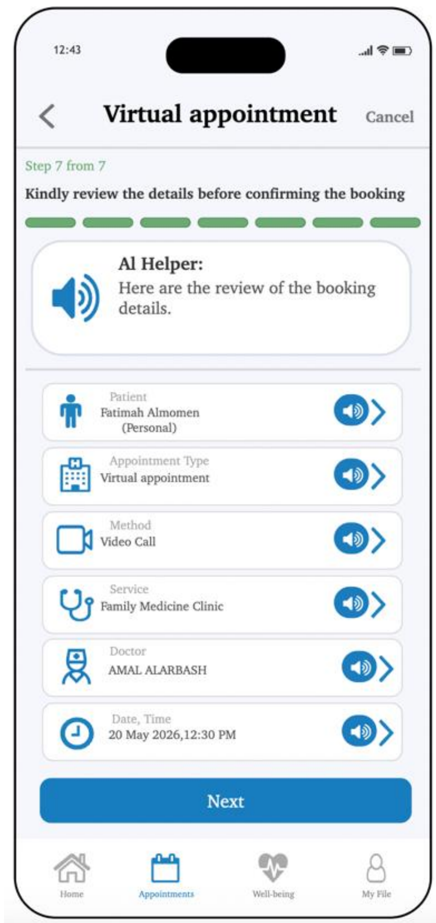

Figure 21. Review Virtual Appointment Details Interface

1.21. Confirm Virtual Appointment Interface

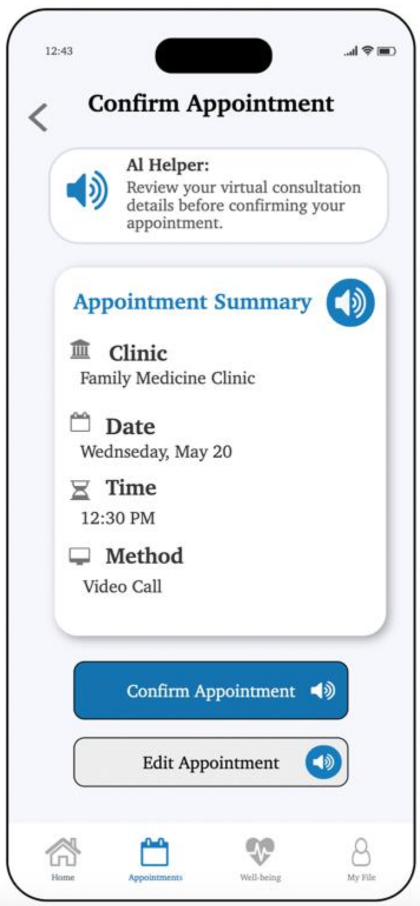

Figure 22. Confirm Virtual Appointment Booking Interface

1.22. Successful Booking Interface

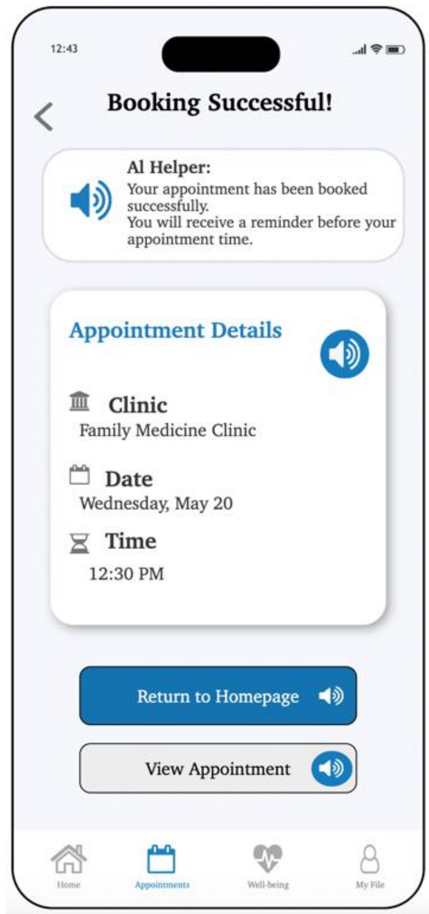

Figure 23. Successful Booking Interface

2. Discussion

In our project we showed how Explainable Artificial Intelligence (XAI) can improve accessibility for visually impaired users in digital healthcare applications. The CarePoint prototype provided clear guidance, audio support, and simple navigation to help users book medical appointments more independently and confidently.

The project also highlighted the importance of applying Human-Computer Interaction (HCI) principles such as reducing cognitive load, maintaining consistent design, and providing clear feedback. These features improve usability and increase user trust in the system. Additionally, the accessibility improvements can benefit a wider range of users, including elderly individuals and people unfamiliar with healthcare applications.

Although the prototype was not tested in a real healthcare environment, the project demonstrates the potential of combining XAI and accessible design to create a more inclusive healthcare experience in Saudi Arabia.

3. Conclusion

In conclusion, this project aims to reduce the heavy dependency that developers and applications have on visual explanation methods in hopes of giving visually impaired individuals a higher sense of independence when navigating technology. By designing high-fidelity prototypes in Microsoft Axure and incorporating AI technologies like Explainable Artificial Intelligence (XAI) into interfaces, we may create a more understandable technological environment that caters not only to those with sight issues but to a wider, more diverse audience as well. Applying HCI principles such as maintaining a consistent, clear design, reducing cognitive load, and increasing trust in automation, in addition to other adjustments such as simplifying process operations to be clearer and more understandable, and utilizing newfound technologies, allows for a prosperous future in the advancement of inclusion of individuals with various disabilities.

Project findings:

Q1. Which explanation style (text, audio, pictorial) best reduces cognitive load and improves comprehension for people with disabilities?
Audio-based explanations appear to be the most effective in reducing cognitive load and improving comprehension for visually impaired users, as they provide clear guidance without relying heavily on visual interaction.

Q2. How do explainable AI interfaces help users with disabilities build accurate mental models of digital services?
Explainable AI interfaces help users with disabilities develop accurate mental models by offering transparent, step-by-step explanations that improve understanding of system processes and navigation.

Q3. How does explainability influence trust in AI guidance among users with disabilities?
Explainability positively influences user trust by providing clear feedback, confirmation messages, and understandable guidance, which increases confidence during interaction with AI-supported systems.

Q4. Can age- and ability-appropriate explainable interfaces reduce confusion and task errors during interaction with digital services?
Yes, age- and ability-appropriate explainable interfaces can reduce confusion and task errors by simplifying navigation, improving accessibility, and providing consistent support throughout the interaction process.

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